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OpenCCM: An Open-Source Continuous Compartmental Modelling Package

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Summary

OpenCCM is a compartmental modelling (Jourdan et al., 2019) software package which is based on recently-developed fully automated flow alignment compartmentalization methods (Vasile et al., 2024). It is primarily intended for large-scale flow-based processes where there is weak coupling between composition changes, e.g. through (bio)chemical reactions, and convective mass transport in the system. Compartmental modelling is an important approach used to developed reduced-order models (Chinesta et al., 2017) (Benner et al., 2020) using a priori knowledge of process hydrodynamics (Jourdan et al., 2019). Compartmental modelling methods, such as those implemented in OpenCCM, enable simulations of these processes to be performed with far less computational complexity while still capturing the key aspects of their performance.

OpenCCM integrates with two multiphysics simulation software packages, OpenCMP (Monte et al., 2022) and OpenFOAM (Greenshields, 2024), allowing for ease of transferring simulation data for compartmentalization. Additionally, it provides users with built-in functionality for calculating residence times, exporting to transfer data to simulation software, and for exporting results for visualization using ParaView (Ayachit, 2015). Post-processing methods are included for mapping simulation results from compartment domains to the original simulation domain, useful for both visualization purposes and for further simulations in other software (e.g. multi-scale modelling).

Statement of Need

Simulation-based design and analysis continues to be widely applied in research and development of physicochemical processes. Processes with large differences in characteristic time- and length-scale result in the infeasibility of direct multiphysics simulations due to computational limitations. This imposes a significant computation costs which severely reduces the utility of these simulations for entire classes of processes. Compartmental modelling is well-suited for such applications because it produces reduced-order models which are orders-of-magnitude less computationally demanding than direct simulation by taking advantage of the weak coupling between the short and long time-scale phenomena.

However, there are several barriers to the more widespread use of compartmental models. The largest of these is the lack of software for automatically generating compartmental models. There exists closed-source software, specifically AMBER (Quintessa, n.d.), for manually creating and solving well-mixed compartment networks. However, this software is cost-prohibitive for much of the research community and lacks automated compartmentalization. There also exists open-source software, Cantera (Goodwin et al., 2023), for solving compartment

networks. However, it does not incorporate flow information, when available, either from direct observation or multiphysics simulation. Furthermore, neither of these software allow for usage and direct transfer of flow information from multiphysics simulations, such as computational fluid dynamics (CFD) simulations, which are typically feasible over short times-scales.

The overall aim of `OpenCCM` is to fill this need for an open-source compartmental modelling package which is user-friendly, compatible with a variety of simulation package back-ends (e.g. `OpenFOAM` and `OpenCMP`), and which fits into the users existing simulation and post-processing software toolchain, i.e. `ParaView`.

Features

FEATURE	DESCRIPTION
Model support	Accepts `OpenCMP` (Monte et al., 2022) and `OpenFOAM` (Greenshields, 2024) results
Compartmentalization	Single-phase flow-based compartment identification
Compartmental Modelling	Plug Flow Reactors (PFRs)-in-series-based model Previous state-of-the-art Continous Stirred-tank reactor (CSTR)-based models
CM Simulations	Linear, non-linear, and reversible arbitrary reactions. 1st Order upwinding finite-difference-based Adaptive time-stepping
Post-Processing	Residence time distribution
Output	Intermediary mesh format Labeled compartments in `Paraview` format Concentrations from CM simulations in both `Paraview` and simulation package format
Performance	Multi-threading Caching of intermediary results to speed-up subsequent runs

User Interface

The `OpenCCM` Python package can be used via text-based configuration files centered around the CLI (command line interface) where each simulation run/project is in a self-contained directory. In addition to the `OpenCCM` configuration files, the required contents include flow information from one of two open source simulation packages: `OpenCMP` and `OpenFOAM`. For `OpenCMP` three files are required:

1) the `OpenCMP` config file,
2) the mesh on which the simulation was run, and
3) the .sol solution file that contains the velocity profile to use for creating the compartmental model.

For `OpenFOAM`, two directories are required:

1) the `constant` directory which contains the mesh information in ASCII format,

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2) a directory containing the simulation results to be used for creating the compartmental model saved in ASCII format.

The path to the solution directory is specified in the `OpenCCM` config, and the `constant` directory is assumed to be in the same parent folder. The `OpenCCM` software will create several output directories: `log/` which contains detailed debugging information (if enabled), `cache/` which contains intermediary files, and `output_ccm/` both simulation results of the compartmental model (in various user-specified formats) and `ParaView` files for visualization.

A sample config file, `CONFIG`, which outlines the available parameters is included in the main directory. An excerpt of it, showing the compartmental modelling parameters, is shown below. Square brackets indicates values which have a default with the default provided inside the brackets.

```
[COMPARTMENT MODELLING]
# Whether to use PFRs-in-series or a CSTR to model each compartment.
model                   = PFR
# Volumetric flow through a single facet
# below which the flow is considered 0.
flow_threshold_facet    = [1e-15]
# Volumetric flow threshold through a surface
# below which the flow is considered 0.
flow_threshold          = [1e-15]
# Maximum allowable difference (in % of compartment volume)
# between connections for merging them to one location.
dist_threshold          = [5 / 100]
# Absolute tolerances for checking that conservation of mass
# after the flow optimization is performed.
atol_opt                = [1e-2]
```

Reaction Configuration File

The chemical reaction parser in `OpenCCM` reads and parses the reactions configuration files and can handle general reaction equations of the form,

```
aA + bB + [...] -> cC + dD + [...]
```

with associated numeric rate constants. It intentionally does not support the standard <-> symbol for reversible chemical reactions, so that each independent reaction has its rate constant explicit defined. Therefore, a reversible reaction must be written as two independent forward reactions, each with its own rate constants. Each species *label* can contain letter and numbers, but cannot contain brackets "(" or ")" or special characters, e.g. "+", "-", "^", etc. Kinetic rate constants must be expressed as positive real numbers in standard or scientific notation. Additionally, each reaction/rate pair must have a unique *identifier* (i.e. R1, R2). For example, take the reversible reaction,

$$2\mathrm{NaCl} + \mathrm{CaCO_3} \Leftrightarrow \mathrm{Na_2CO_3} + \mathrm{CaCl_2}$$

with $k_f = 5e-2$ and $k_r = 2$, dimensionless numbers picked for demonstration purposes. A configuration file for this reversible reaction may then be:

```
[REACTIONS]
R1: 2NaCl   + CaCO3 ->  Na2CO3 + CaCl2
R2:  Na2CO3 + CaCl2 -> 2NaCl   + CaCO3

[RATES]
R1: 5e-2
```

```
R2: 2
```

where **R1** and **R2** are the reaction *identifiers* for the forward and reverse reactions, respectively. An example reaction config file, `CONFIG_REACTIONS` is provided in the main directory.

Examples of Usage

Several examples are provided in the `OpenCCM` documentation which demonstrate the usage of both `OpenCMP` and `OpenFOAM` simulation flow information for compartmentalization. One example, inside `examples/OpenCMP/pipe_with_recird_2d`, uses the geometry from (Vasile et al., 2024) and shows how to execute the needed CFD simulation for flow information (both using `OpenCMP` and `OpenFOAM`), create/visualize the compartmental model results, and compare the predicted RTD to the reference result directly from CFD simulation.

For this illustrative example, the steady-state hydrodynamic flow-profile is obtained by running the `OpenCMP` simulation through the `run_OpenCMP.py` script in the folder. The resulting flow profile was opened in `ParaView` and the line integral convolution of the velocity field is shown below, colored by velocity magnitude.

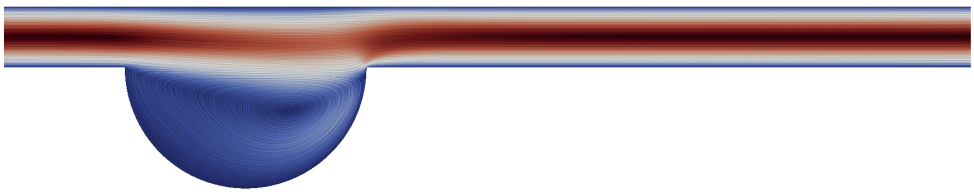

Figure 1: Visualization of hydrodynamics from CFD simulation with line integral convolutions indicating local flow direction and color corresponding to velocity magnitude.

The underlying velocity field data is then processed using `OpenCCM` to produce a network of compartments by executing the `run_compartment.py` script. The figure below shows each element of the original mesh colored according to the compartment it belongs to.

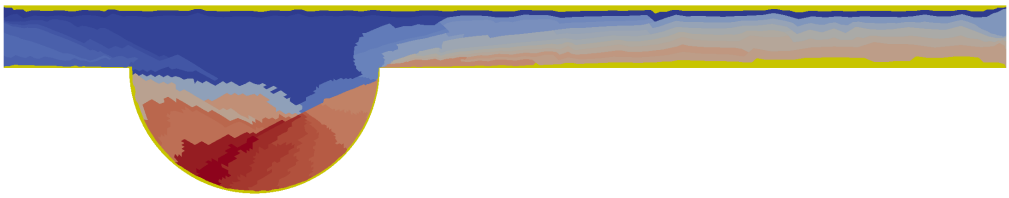

Figure 2: Visualization of flow-informed compartmentalization with coloring corresponding to compartment number.

That network of compartments is further processed as each compartment is represented by a series of plug flow reactors (PFRs). The resulting network (graph) of PFRs is shown in the figure below; nodes are the centers of the PFRs and edges are connections (flows) between PFRs.

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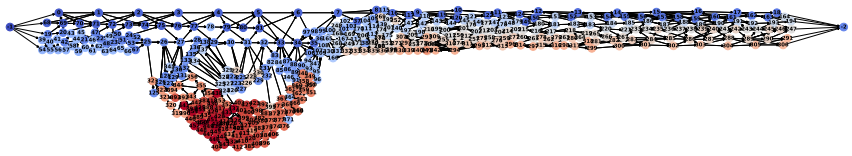

Figure 3: Undirectly graph of the compartment network resulting from both (i) flow-information compartmentalization and (ii) the use of spatially-varying compartment approximations (PFRs).

RTD Curves

The Residence Time Distribution (RTD) curve for both the CFD and Compartmental Model (CM) are calculated using the script in the supplementary material of (Vasile et al., 2024).

Residence time distribution curves for CFD and CM simulations.

Figure 4: Residence time distribution curves for CFD and CM simulations.

Reactions

Finally, to demonstrate how to use the reaction system we will implement the reversible reaction system mentioned above:

$$2\mathrm{NaCl} + \mathrm{CaCO_3} \Leftrightarrow \mathrm{Na_2CO_3} + \mathrm{CaCl_2}$$

with $k_f = 5e-2$ and $k_r = 2$ as the dimensionless forward and backwards rate constants whose values were picked for demonstration purposes. All species start at an initial concentration of 0, dimensionless concentration, and have inlet boundary conditions of $[\mathrm{NaCl}] = [\mathrm{CaCO_3}] = 1$ and $[\mathrm{Na_2CO_3}] = [\mathrm{CaCl_2}] = 0$, again both dimensionless concentration. The equations and conditions have already been specified, and the simulation can be run by using the `run_compartment_w_rxn.py` script. Note that when you re-run this script it will finish much faster than the first time since the compartmental model is loaded from disk rather than having to be re-created each time.

To analyze the results, the equilibrium values for this reversible system are calculated as follows:

$$k_f[\mathrm{NaCl}]^2[\mathrm{CaCO_3}] = k_r[\mathrm{Na_2CO_3}][\mathrm{CaCl_2}]$$

$$\frac{k_f}{k_r} = \frac{[\mathrm{Na_2CO_3}][\mathrm{CaCl_2}]}{[\mathrm{NaCl}]^2[\mathrm{CaCO_3}]}$$

$$\frac{5 \times 10^{-2}}{2} = \frac{(x)(x)}{(1-2x)^2(1-x)}$$

$$x \approx 0.1147$$

where x is the change in $\mathrm{CaCO_3}$, in dimensionless units.

The expected equilibrium concentrations for the four species are: $[NaCl] = 0.7706$, $[CaCO_3] = 0.8853$, $[Na_2CO_3] = 0.1147$, and $[CaCl_2] = 0.1147$. Based on the figures below, and from opening up the results, it can be seen that these steady state values are obtained at the outlet of the reactor.

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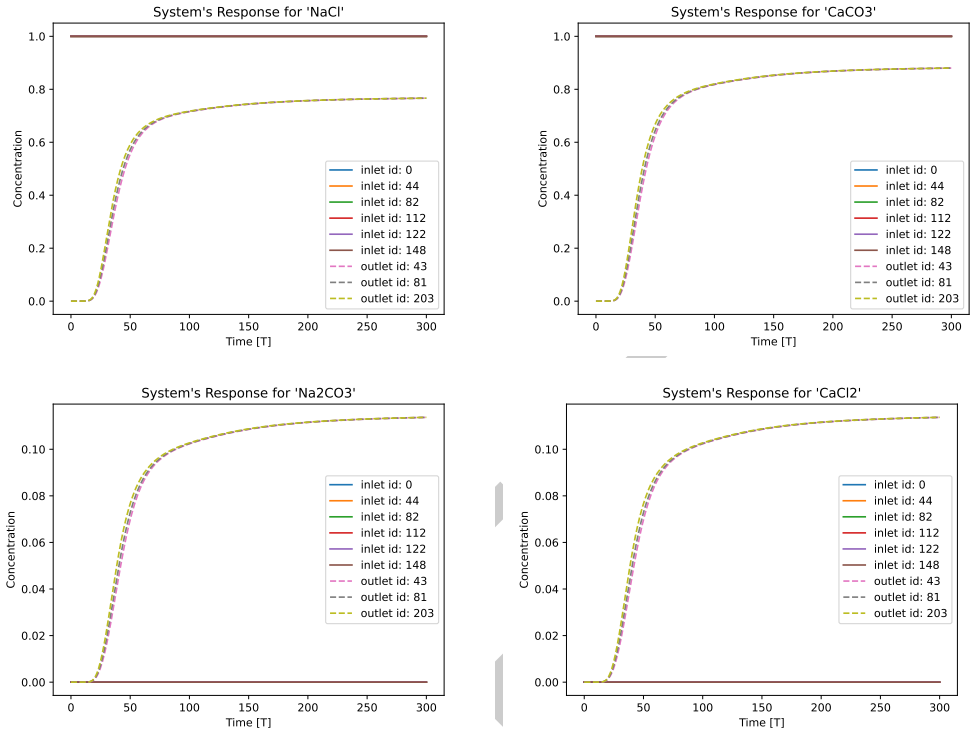

To output the `ParaView` visualizations, change `output_VTK` to `True` in the `CONFIG` file and re-run the simulation.

Acknowledgements

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